Unattended SAP GUI Automation

From Manual ERP Processing to Fully Unattended Execution

Business Context

- High-volume SAP material processing
- Manual multi-screen ERP workflow
- Operational inefficiency and risk

Technical Constraints

- No API access – SAP GUI scripting only
- Excel used as orchestration layer
- OLE / COM interruptions blocking execution

Core Challenge – OLE Dialog

- OLE warnings interrupted SAP ↔ Excel interaction
- Manual confirmation required
- Prevented true unattended automation

Solution Implementation

- Identified dialog trigger points
- Implemented alert suppression
- Added defensive validation
- Achieved fully unattended execution

Before vs After Automation

BEFORE

- Manual navigation
- 5–8 min per record
- OLE interruptions
- Human-in-the-loop
- Multi-day processing

AFTER

- Fully unattended execution
- Overnight runtime
- OLE suppression
- Deterministic output
- Scalable & restartable

Quantified Impact

Manual: ~5–8 minutes per record

- 1,000 records ≈ 80–130 hours manual work
- Automation: Overnight unattended processing
- Reduced operational risk and human error

Skills Demonstrated

- ERP workflow analysis (SAP GUI)
- Unattended automation architecture design
- COM / OLE exception handling
- Task queue & state machine logic
- Operational reliability engineering

Future Scalability / Roadmap

Phase 1 – Stabilization: Harden exception handling & retry logic

Phase 2 – Observability: Add structured logging, audit trails & metrics

Phase 3 – Orchestration: Integrate with RPA platform (UiPath / Power Automate)

Phase 4 – API Transition: Replace GUI steps with API calls where available

Phase 5 – Scalability: Parallel execution & workload distribution

Risk & Mitigation Strategy

Risk 1: SAP GUI instability or session timeout

- Mitigation: Session validation & controlled reinitialization

Risk 2: OLE / COM dialog reoccurrence

- Mitigation: Alert suppression + boundary control logic

Risk 3: Data inconsistency or partial extraction

- Mitigation: Validation layer & retry mechanism

Risk 4: Long runtime & scaling limits

- Mitigation: Roadmap toward orchestration & API transition

Lessons Learned

- True automation is achieved only when human intervention is eliminated
- Exception handling is more critical than core logic
- GUI automation requires system-level thinking
- Divide & Conquer improves maintainability and resilience
- Designing for restartability increases operational trust